ENVS-193DS-spring2026-Final

Mila Matich

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Table of contents

Here is the link to my GitHub repository.

Set Up

```
# loading in packages for final
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file organization
library(rstatix) # calculate cohen's d
library(MuMIn) # model selection
library(DHARMa) # to simulate residuals
library(ggeffects) # getting model predictions

# reading in data and storing as objects
nests <- read_csv(here("data","nest_data_final.csv"))
personal <- read_csv(here("data","personal_data.csv"))

# cleaning personal data for problem 3
personal_clean <- personal |>
  # cleaning column names
  clean_names() |>
  # selecting columns of interest
  select(weekday_or_weekend,
         step_count)
```

Problem 1: Research Writing

a. Transparent statistical methods

In part 1, they used a generalized linear model (GLM) to show how distance to water edge predicts the probability of American Avocet microhabitat use. **In part 2, they used a chi-squared test** to compare counts between bird species across different habitat sites.

b. Suggestions for rewriting

Part 1:

Habitat distance to water edge was a significant predictor of whether or not the shorebird American Avocet used the microhabitat.

Part 2:

Our study shows that the distribution of bird species differed significantly across different habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat). These results suggest that the studied bird species (American Avocet, Forster's Tern, Black-necked Stilt) use habitats differently, and may prefer some habitats more than others.

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use could be seasonality measured across a long time series and keeping track of date (categorical variable) that would be categorically placed into Fall, Spring, Summer or Winter. This factor would influence the habitat use as it influences shorebird migration patterns, which determines whether or not the birds are in the location of study, and therefor keeping track of this data alongside the known migration pattern of each species will help explain why birds may be present at different times.

Problem 2: Data Analysis

a. Clean and wrangle

```
# creating a new object called: nests_clean for cleaned data set
nests_clean <- nests |>
  # creating new columns to store full scientific species name
  mutate(ant_species = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti", # C. sjostedti
    "RRB" ~ "Crematogaster mimosae", # C. mimosae
    "TP" ~ "Tetraponera penzigi", # T. penzigi
    "BBR" ~ "Crematogaster nigriceps", # C. nigriceps
  )) |>
  # filter to only include: C. mimosae, C. sjostedti, and C. nigriceps
  filter(ant_species %in% c("Crematogaster mimosae",
```

```r
                             "Crematogaster sjostedti",
                             "Crematogaster nigriceps")) |>
  # reordering ant species levels according to instructions
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                    "Crematogaster sjostedti",
                                    "Crematogaster mimosae",
                                    "Crematogaster nigriceps")) |>
  # selecting columns of interest for the research questions (only 3)
  select(height_cm, ant_species, case_control)

# display the structure of the new data object
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ ant_species : Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
```

```r
# display 10 random rows
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 3
   height_cm ant_species             case_control
       <dbl> <fct>                          <dbl>
 1      150. Crematogaster mimosae              0
 2      178  Crematogaster mimosae              0
 3      130  Crematogaster mimosae              0
 4      456  Crematogaster sjostedti            0
 5      144  Crematogaster mimosae              0
 6      169  Crematogaster mimosae              0
 7      166. Crematogaster nigriceps            1
 8      146. Crematogaster mimosae              0
 9      142  Crematogaster mimosae              0
10      164  Crematogaster mimosae              0
```

b. Response variable

The 1s and the 0s in a biological context represents a binary outcome of presence (1) or not present (0). In this specific study, a 1 represents a tree with a nest and 0 represents a tree with no nests.

c. Purpose of study

One of the predictors is tree height, which is a numerical, continuous variable (measured in cm). As tree height increases, the probability of nest occurrence would increase, as determined by the the logistic model in the results section (paragraph 3), however the results also suggest that there is no relationship between height and nest probability (paragraph 3). This is likely because height can shield the nests from ground predators, however this is dependent on what ant species are present as it can alter the architecture of the tree (ie. the denser canopy when *C. nigriceps* is present), as inferred from the discussion (paragraph 2).

The other predictor is acacia ant species, which is a categorical variable (differentiated by type of ant: *C. mimosae*, *C. sjostedti*, and *C. nigriceps*). Because ants are nest predators in acacia trees, it is predicted that the number of nests will be higher in less aggressive ant-tree symbionts (*C. sjostedti*), however, if the acacia ants defend the tree from other nest predators, it is predicted that number of nests will be higher in the more aggressive symbionts (*C. mimosae* and *C. nigriceps*). This information is found in the third to fourth paragraph in the introduction.

d. Table of models

Model number	Ant Species	Tree Height	Model Description
1	X	X	Both terms w/o interaction (+)
2	X	X	Both terms w/ interaction (*)

e. Run the models

```
# running model 1 (no interaction)
nest_mod1 <- glm(case_control ~ ant_species + height_cm, # + = no interaction
                 data = nests_clean, # from cleaned data set
                 family = "binomial") # binomial data set

# running model 2 (interaction)
nest_mod2 <- glm(case_control ~ ant_species * height_cm, # * = interaction
                data = nests_clean, # from cleaned data set
                family = "binomial") # binomial data set
```

f. Select the best model

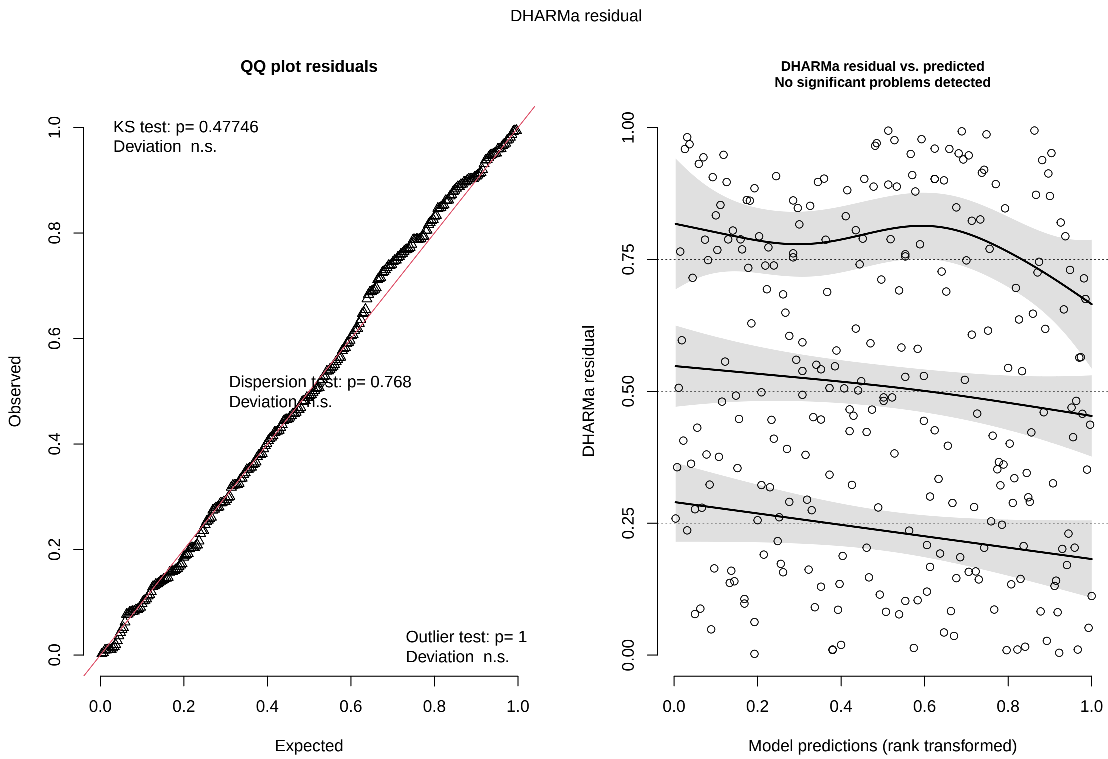
```
# calculating AIC for the two models
AICc(nest_mod1,
     nest_mod2) |>
  # rearranging output to be in order of lowest to highest AIC value
  arrange(AICc)
```

```
          df     AICc
nest_mod2  6 238.1236
nest_mod1  4 258.8940
```

The best model for predicting the probability of bird nest presence as determined by Akaike's Information Criterion (AIC) is the model of interaction between tree height and ant species as it has the lower AIC.

g. Check the diagnostics

```
# running diagnostic plots using stimulated residuals from DHARMa package
plot(
  simulateResiduals(nest_mod2) # using interaction model (2)
)
```



There is a uniform distribution of the data as shown by the QQ residuals plot (the data points tightly follow the QQ reference line). On the residuals vs. predicted plots, there are no clear pattern of data points, and they are randomly scattered across plots, thus there are randomly distributed residuals. In ocnclusion, the assumptions to use a generalized linear model are met.

h. Visualize the models predictions

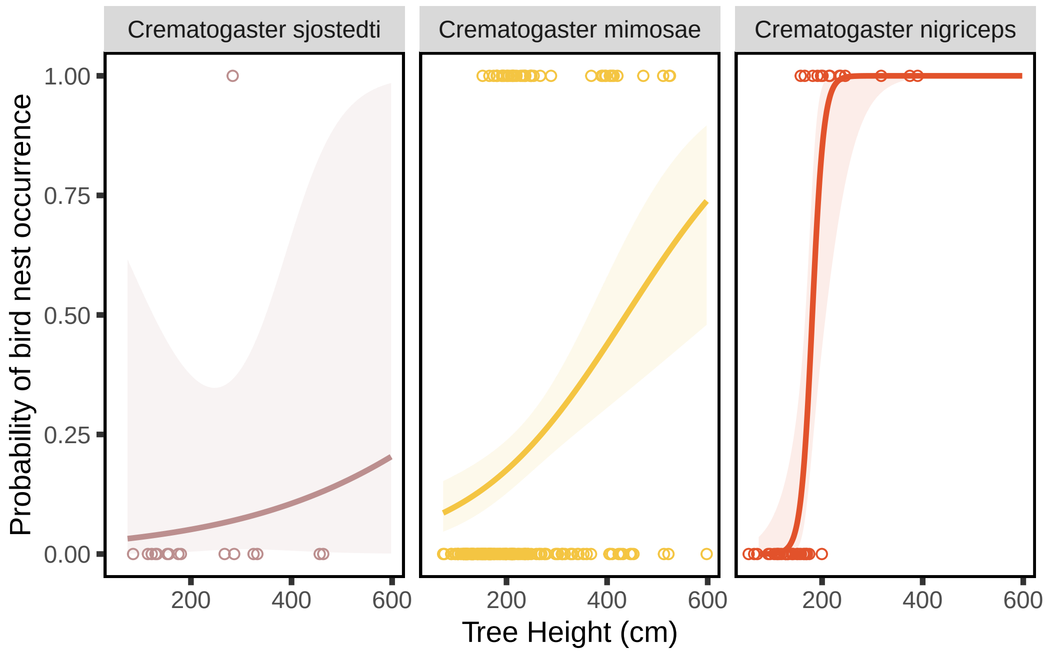
Generating model predictions first

```
# generating a object for model 2 prediction
preds <- ggpredict(
  nest_mod2, # from model 2 object
  # terms = height (its range) & ant species
  terms = c("height_cm [74:598 by = 1]", "ant_species")) |>
  # renaming columns so that it makes it easier to build the figure
  rename(height_cm = x, # x --> height_cm
         ant_species = group) # groups --> ant_species
```

Making the Figure

```
# base layer: ggplot
ggplot(data = nests_clean, # data is from cleaned data set
       mapping = aes(x = height_cm, # x-axis: tree height
                     y = case_control)) + # y-axis: nest probability
  # first layer: geom points (show underlying data)
  geom_point(mapping = aes(color = ant_species), # color based on species
             shape = 21) + # shape of points are circles
  # second layer: geom_line (based on predictions)
  geom_line(data = preds, # data from model predictions
            mapping = aes(x = height_cm, # x-axis: tree height
                          y = predicted, # y-axis: predicted probability
                          color = ant_species), # color based on species
            linewidth = 1) + # setting line width
  # third layer: ribbon (95% confidence interval predictions)
  geom_ribbon(data = preds, # data from model predictions
              mapping = aes(x = height_cm, # x-axis: tree height
                            y = predicted, # y-axis: predicted probability
                            ymin = conf.low,
                            ymax = conf.high,
                            fill = ant_species), # color based on species
                          alpha = 0.1) + # opaque points
  # making a new panel for each species
  facet_wrap(~ ant_species) +
  # adding custom colors based on species for the points/line/ribbon
  scale_color_manual(values = c(
    "Crematogaster mimosae" = "#F4C542",
    "Crematogaster sjostedti" = "rosybrown",
    "Crematogaster nigriceps" = "#E2522B"))  +
  # adding custom colors based on species
  scale_fill_manual(values = c(
    "Crematogaster mimosae" = "#F4C542",
    "Crematogaster sjostedti" = "rosybrown",
    "Crematogaster nigriceps" = "#E2522B")) +
  # labeling x and y axis
  labs(x = "Tree Height (cm)",
       y = "Probability of bird nest occurrence") +
  # editing theme
  theme(panel.background = element_rect(fill = "white", # no panel background
                                        colour = "black", # black outline
                                        linewidth = 1), # width of outline
```

```
panel.grid = element_blank(), # getting rid of gridlines
axis.ticks = element_line(linewidth = 1), # adding tick marks on axes
legend.position = "none") # getting rid of legend
```



i. Write a caption for your figure

Figure 1. Probability of bird nest occurrence differs based on tree height and the present ant species. Figures are separated by ant species and this is differentiated through distcitive panels and colors, where *C. sjostedti* is brown, *C. mimosae* is yellow, and *C. nigriceps* is red. Open circle points represent individual observations of nest occurrence in each scenario, the line represents model predictions of nest occurrence given tree height (cm) and ant species, and the ribbon represents the 95% confidence interval. Data originally from Lujan, Ema et al. 2023. "Symbiotic acacia ants drive nesting behavior by birds in an African savanna." Biotropica 55:6. doi: 10.1111/btp.13276.

j. Calculate model predictions

```
# running ggpredict for model predictions @ height = 600cm
ggpredict(nest_mod2, # taking data from interaction model (2)
```

```
          terms = c("height_cm [600]", # 1st term = height @ 600cm
                    "ant_species")) # 2nd term = ant species
```

```
# Predicted probabilities of case_control

ant_species: Crematogaster sjostedti

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species: Crematogaster mimosae

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: Crematogaster nigriceps

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

The predicted probability of nest occurrence at the tallest tree height (600 cm) with *C. sjostedti* present is 0.2 (95% CI: [0.00, 0.99]), with *C. mimosae* present is 0.74 (95% CI: [0.48, 0.90]), and *C. nigriceps* present is 1.00 (95% CI: [1.00, 1.00]). According to these predictions and Figure 1, birds preferred to nest in trees inhabited by *C. mimosae* and *C. nigriceps*, with more nesting on the taller trees with both of these species, however this interaction is more distinctive with trees inhabited by *C. nigriceps*. Birds also rarely nested on trees with *C. sjostedti* present, regardless of height.

Birds have a selection preference towards trees inhabited by more aggressive ant species (*C. mimosae* and *C. nigriceps*) as they do not directly attack the nestlings or adult birds and may reduce the risk of predation by vertebrate and invertebrate herbivores. The specific preference toward *C. nigriceps* out of the two aggressive species is due to tha fact that when this species it present, it alters the trees shoot structure leading to denser canopies, which further protects the nests from other predators such as raptors or snakes.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

- My affective visualizations differ greatly from my exploratory visualization in Homework 2 as it represents my data in a different way. Instead of using a linear model or a box plot, I use a bar graph diagram in order to show my step count on each given day. It also shows how my data fluctuates through time as each bar represents one day, while the visualizations in homework 2 do not differentiate each day on a linear time scale (they are just represented by points). Additionally, instead of continuing to explore temperature as a predictor variable, I chose to do amount of homework as I felt like it would be easiest to visualize.
- They are similar in that both visualizations show each individual data point, where this is through points in both of my graphs from Homework 2, and done by bars in my affective visualization.
- Some patterns that I see is that step count was higher visually on weekends compared to weekdays (as shown by the median bar on the boxplot and the height of the outlined bars in my affective visualization). Additionally, like how temperature affected my step count, there seems to not be a pattern in homework levels and step count.
- I got a lot of feedback for my color scheme and also for my artist statement. I was told that my color scheme was misleading because while brown represented the "medium" amount of homework, it stood at the most and should have represented the "high" amount of homework, and that I should stick to a classic high, medium, low color scheme with red purple and blue, or keep it monochromatic with different shades. That being said, I changed my colors to be gray, yellow, and red to show low, medium, high as it makes more sense visually and intuitively. I also got feedback on my artist statement to include more information about what my piece is showing and why, which I ended up including.

b. Designing an analysis

The main question that I was asking, regardless of the other variables I measured, was how my step count differed between the days that I had class and days that I did not (ie. weekday vs. weekend). That being said, my predictor variable would be day of the week, as a binary, categorical variable (weekday vs. weekend) and my response variable would be step count, which is a discrete numerical variable.

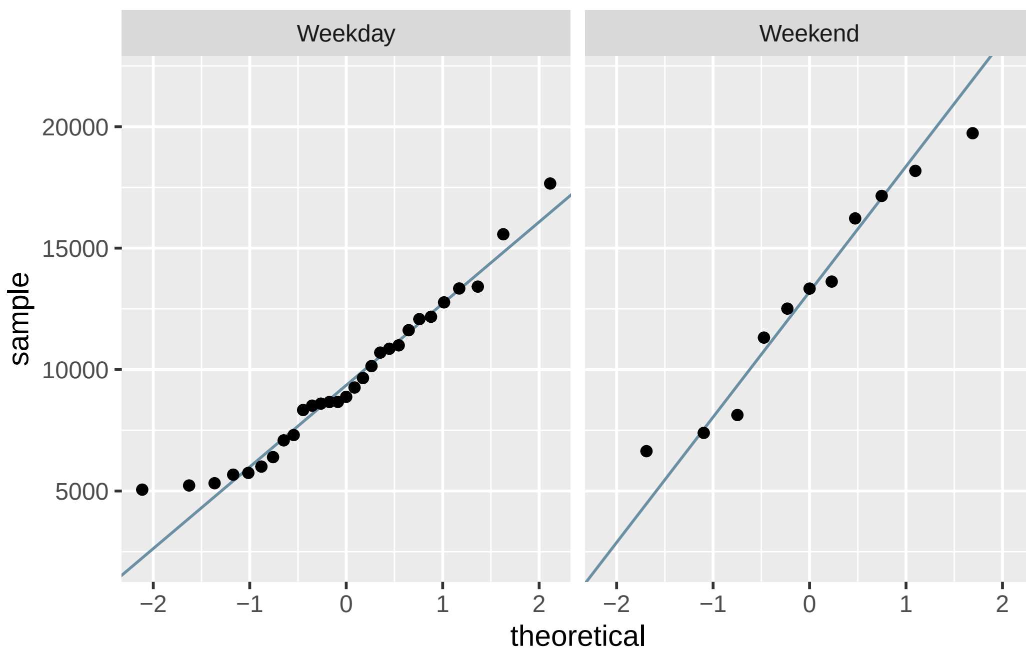
An analysis I could run to answer this question would be Two-Sample T-test, however the specific type of test depends on whether my data is normal and variances are equal (Student's if variances are equal and data is normally distributed, Welch's if variances are not equal but data is normally distributed, and Mann Whitney U if data are not normally distributed). This test makes sense as it will compare my average (or median) step count across the weekdays and the weekends, which is two groups of data. It also make sense because my predictor variable is categorical, and I want to compare step count across two groups and see how they differ.

c. Check any assumptions and run your analysis

Checking Assumptions

Normality

```
# base layer: gg-plot
ggplot(data = personal_clean, # using cleaned personal data frame
       # y-axis is step count
       mapping = aes(sample = step_count)) +
  # first layer: QQ reference line
  geom_qq_line(color = "#6B8FA3") + # coloring the line
  # second layer: adding qq points
  geom_qq() +
  # creating new panels for each treatment type
  facet_wrap(~ weekday_or_weekend)
```



Data points on both weekday and weekend QQ plots roughly follow the reference line, besides a bit of deviation on the tails. Thus, it can be concluded that the data for weekend and weekday are normally distributed and the assumption of normality is met.

Variances

```
# checking variances using var.test
var.test(
  # formula: response variable ~ grouping variable
  step_count ~ weekday_or_weekend,
  # data from cleaned data set
  data = personal_clean
)
```

```
    F test to compare two variances

data:  step_count by weekday_or_weekend
F = 0.52113, num df = 28, denom df = 10, p-value = 0.17
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1566534 1.3274935
sample estimates:
ratio of variances
         0.5211343
```

We determined that group variances are equal (F-test, $F(28, 11) = 0.521$ $p = 0.17$).

Running Analysis

Student's T-Test

```
# running a Student's t-test using results from normality & variance test
t.test(
  # formula: response variable ~ grouping variable
  step_count ~ weekday_or_weekend,
  # adding the argument for equal variances
  var.equal = TRUE,
  # using data from cleaned personal data frame
  data = personal_clean
)
```

```
	Two Sample t-test

data:  step_count by weekday_or_weekend
t = -2.8381, df = 38, p-value = 0.007245
alternative hypothesis: true difference in means between group Weekday and group Weekend is 
95 percent confidence interval:
 -6176.590 -1033.573
sample estimates:
mean in group Weekday mean in group Weekend 
             9506.828             13111.909 
```

Effect Size

```r
# calculating the effect size between step count and weekday/weekend
# formula: response variable ~ grouping variable
cohens_d(step_count ~ weekday_or_weekend,
         # pulling from personal clean data set
         data = personal_clean)
```

```
# A tibble: 1 x 7
  .y.        group1  group2  effsize    n1    n2 magnitude
* <chr>      <chr>   <chr>     <dbl> <int> <int> <ord>    
1 step_count Weekday Weekend  -0.927    29    11 large    
```

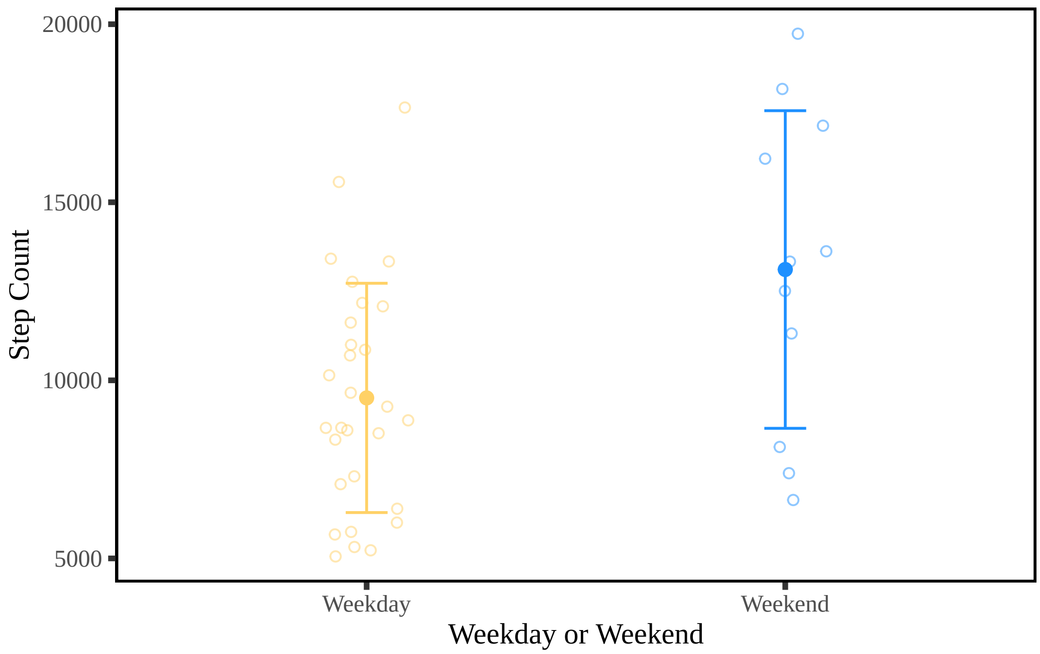
d. Create a visualization

Generating Summary for Jitter Plot

```r
# creating a new object called urchins_summary to calculate values
personal_summary <- personal_clean |>
  # group by weekday or weekend
  group_by(weekday_or_weekend) |>
  # calculating values for each category
  summarize(mean = mean(step_count), # calculating mean
            n = length(step_count), # calculating sample size
            sd = sd(step_count)) # calculating standard deviation
```

Generating Jitter Plot for T-Test

```
# base layer: ggplot
ggplot(data = personal_clean,
       mapping = aes(x = weekday_or_weekend, # x-axis = weekday/weekend
                     y = step_count,# y-axis = step count
                     color = weekday_or_weekend)) + # assigning colors
  # first layer: adding underlying data (individual observations)
  geom_jitter(width = 0.1, # narrow jitter width
              height = 0, # no vertical jitter/point movement along the y-axis
              alpha = 0.5, # transparent points to be visually distinct
              shape = 21) + # open circle points
  # second layer: adding point for the mean
  geom_point(data = personal_summary, # data from summary
             mapping = aes(x = weekday_or_weekend, # x-axis = weekday/weekend
                           y = mean), # y-axis = mean value of steps
             size = 2) + # adjusting point size
  # third layer: adding error bars to show variation
  geom_errorbar(data = personal_summary, # data from summary
                mapping = aes(x = weekday_or_weekend, # x-axis = weekday/weekend
                              y = mean, # y-axis = mean
                              ymin = mean - sd, # min y-value on error bar
                              ymax = mean + sd), # max y-value on error bar
                width = 0.1) + # make bars narrower
  # selecting colors for each day category
  scale_color_manual(values =c("Weekday" = "#FFD166", # Weekday = yellow
                               "Weekend" = "dodgerblue")) + # Weekend = blue
  # labeling x and y axis
  labs(x = "Weekday or Weekend", # x labeled as "Weekday or Weekend"
       y = "Step Count") + # y labeled as "Step Count"
  # changing theme elements
  theme(text = element_text(family = "serif"), # font = serif
        panel.background = element_rect(fill = "white", # no panel background
                                        colour = "black", # black outline
                                        linewidth = 1), # width of outline
        panel.grid = element_blank(), # getting rid of gridlines
        axis.ticks = element_line(linewidth = 1), # adding tick marks on axes
        legend.position = "none") # getting rid of legend
```



e. Write a caption

Figure 2. Number of steps taken differs on weekday versus weekend. Small circular points represent the total number of steps recorded on a given weekday (yellow points) or weekend (blue points) from 2026-04-21 to 2026-05-30. The average step count is shown by the solid dot, while the error bars represent the standard deviation of the data to show the variability within the weekdays or weekends. Data collected by Mila Matich from 2026-04-21 to 2026-05-30.

f. Write about your results

My results suggest that my step count was greater on weekends (mean = 13,111.91 steps) than weekdays (mean = 9,506.83 steps). I found a large negative effect (Cohen's d = -0.927) of day-type on mean step count between the weekdays (n = 29) and weekends (n = 11) (Student's t-test, $t(38) = -2.8381$, $p = 0.007245$, $\alpha = 0.05$).

g. Sharing your affective visualization